

Non-Canonical Prognostic DEGs × Immune Landscape: Cross-Phase Biomarker Integration

CACNA1A (HR=0.58, C-index=0.674): best discriminating biomarker in OSCC — does low CACNA1A also mark coldest TME?

PITX2 (HR=1.29): top adverse transcription factor co-activated with NRF2 — does high PITX2 correlate with Tregs?

GATA3 (HR=0.77): epithelial lineage enforcer lost in NRF2-High — does GATA3 loss track with immune exclusion?

Establishing multi-level hierarchy: Redox → Immune → Clinical outcome

